

TASK 1

Task 1(a): Write a Java program to perform encryption and decryption using the algorithm Ceaser Cipher.

Aim: To write a Program on encryption and decryption using ceaser cipher.

Program:

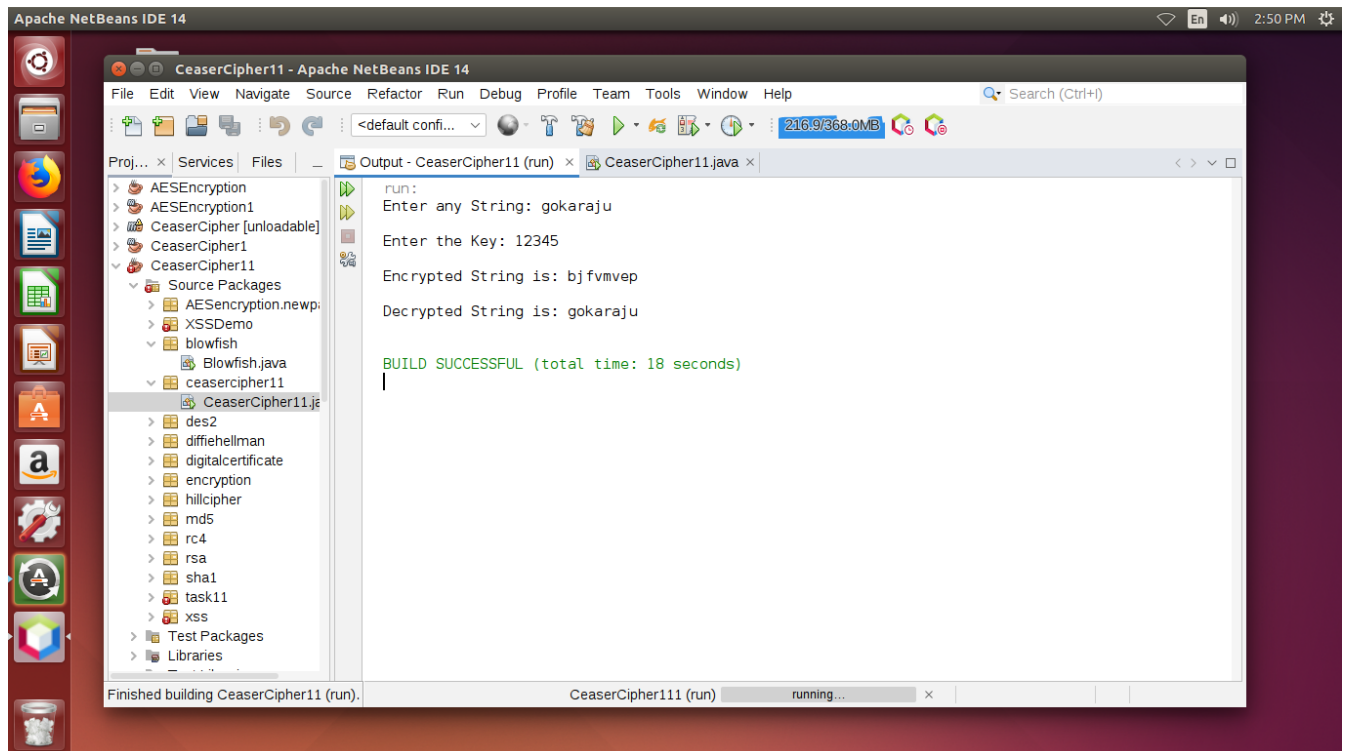
```
import java.security.*;
import java.io.*;
import java.util.*;
//Ceaser Cipher
public class Task1a {
    static Scanner s = new Scanner(System.in);

    private static String encrypt(String data, int key) {
        StringBuilder res = new StringBuilder();
        for (Character ch :
            data.toCharArray()) { if
            (Character.isUpperCase(ch)) {
                char shiftedChar = (char) (((ch - 'A' + key) % 26) +
                    'A'); res.append(shiftedChar);
            } else if (Character.isLowerCase(ch)) {
                char shiftedChar = (char) (((ch - 'a' + key) % 26) +
                    'a'); res.append(shiftedChar);
            } else {
                res.append(ch);
            }
        }
        return res.toString();
    }

    private static String decrypt(String data, int key)
    { return encrypt(data, 26 - (key % 26));
    }

    public static void main(String[] args) {
        String data = s.nextLine();
        int key = s.nextInt();
        data = encrypt(data, key); // Encryption
        System.out.println("Encryption: " + data);
        System.out.println("Decryption: " + decrypt(data, key)); // Decryption
    }
}
```

OUTPUT:



VIVA QUESTIONS

1. How do you decrypt a message that was encrypted using the Caesar Cipher?
2. What are the limitations of the Caesar Cipher in terms of security?
3. How can the Caesar Cipher be broken using brute force or frequency analysis?
4. What is the fundamental principle behind the Caesar Cipher?

Task 1(b):Write a Java program to perform encryption and decryption using the algorithm Substitution Cipher.

Aim: To write a Java Program using substitution cipher.

Program :

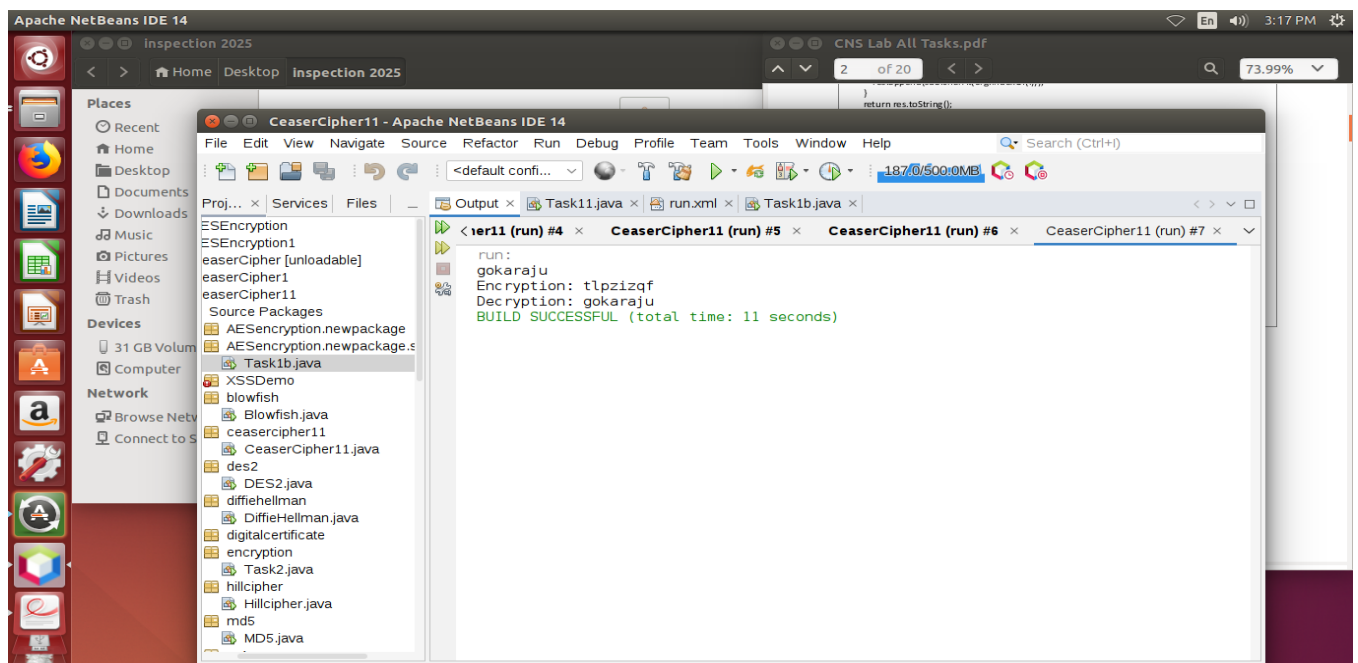
```
package com.islab;
import java.io.*;
import java.util.*;
public class SubstitutionCipher
{
    static Scanner sc = new Scanner(System.in);
    static BufferedReader br = new BufferedReader(new InputStreamReader(System.in));
    public static void main(String[] args) throws IOException
    {
        String encrypt = encrypt();
        String decrypt = decrypt(encrypt);
        System.out.println("The encrypted data is: " +encrypt);
        System.out.println("The decrypted data is: " +decrypt);
    }
    public static String encrypt()throws IOException
    {
        String encrypt = "";
        String a = "abcdefghijklmnopqrstuvwxyz";
        String b = "zyxwvutsrqponmlkjihgfedcba";
        System.out.print("Enter any string: ");
        String str = br.readLine();
        char c;
        for(int i=0;i<str.length();i++)
        {
            c = str.charAt(i);
            int j = a.indexOf(c);
            encrypt = encrypt+b.charAt(j);
        }
        return encrypt;
    }
}
```

```

public static String decrypt(String encrypt){
    String a = "abcdefghijklmnopqrstuvwxyz";
    String b = "zyxwvutsrqponmlkjihgfedcba";
    String decrypt = "";
    char c;
    for(int i=0;i<encrypt.length();i++) {
        c = encrypt.charAt(i);
        int j = a.indexOf(c);
        decrypt = decrypt+b.charAt(j);
    }
    return decrypt;
}
}

```

OUTPUT:



VIVA QUESTIONS

1. What is the Substitution Cipher, and how does it differ from the Caesar Cipher?
2. What are the different types of substitution ciphers? Provide examples.
3. How can a Substitution Cipher be broken using frequency analysis?
4. What are the advantages and disadvantages of using a Substitution Cipher?

Task 1(c):Write a Java program to perform encryption and decryption using the algorithm Hill Cipher.

Aim:To write a Java Program using Hill cipher.

Program:

```
package com.islab;
import java.util.*;
class Basic
{
    String allChar="ABCDEFGHIJKLMNOPQRSTUVWXYZ";
    int indexOfChar(char c)
    {
        for(int i=0;i < allChar.length();i++)
        {
            if(allChar.charAt(i)==c)
                return i;
        }
        return -1;
    }

    char charAtIndex(int pos)
    {
        return allChar.charAt(pos);
    }
}
class Hill
{
    Hill(int block)
    {
        this.block=block;
    }

    Basic b1=new Basic();
    int block=2;
    int key[][]=new int[block][block];

    void keyInsert()throws Exception
    {
        Scanner scn=new Scanner(System.in);
        System.out.println("Enter key Matrix");
        for(int i=0;i < block;i++)
        {
            for(int j=0;j < block;j++)
```

```

        {
            key[i][j]=scn.nextInt();
        }
    }
}

void KeyInverseInsert()throws Exception
{
    Scanner scn=new Scanner(System.in);
    System.out.println("Enter key Inverse Matrix:");
    for(int i=0;i < block;i++)
    {
        for(int j=0;j < block;j++)
        {
            key[i][j]=scn.nextInt();
        }
    }
}

String encryptBlock(String plain)throws Exception
{
    plain=plain.toUpperCase();
    int a[][]=new int[block][1],sum=0;
    int cipherMatrix[][]=new int[block][1];
    String cipher="";
    for(int i=0;i < block;i++)
    {
        a[i][0]=b1.indexOfChar(plain.charAt(i));
    }
    for(int i=0;i < block;i++)
    {
        {
            for(int j=0;j < 1;j++)
            {
                for(int k=0;k < block;k++)
                {
                    sum=sum+key[i][k]*a[k][j];
                }
                cipherMatrix[i][j] = sum%26;
                sum = 0;
            }
        }

        for(int i=0;i < block;i++)
        {
            cipher+=b1.charAtIndex(cipherMatrix[i][0]);
        }
    }
}

```

```

        }
    return cipher;
}
String encrypt(String plainText)throws Exception
{
    String cipherText="";
    keyInsert();
    plainText=plainText.toUpperCase();
    int len=plainText.length();
    // System.out.println(plainText.substring(1,2+1));
    while(len%block!=0)
    {
        plainText+="X";
        System.out.println(len);
        len=plainText.length();
    }
    for(int i=0;i < len-1;i=i+block)
    {
        cipherText+=encryptBlock(plainText.substring(i,i+block));
        cipherText+=" ";
    }
    return cipherText;
}

String decryptBl(String cipher)throws Exception
{
    cipher=cipher.toUpperCase();
    int a[][]=new int[block][1],sum=0;
    int plainMatrix[][]=new int[block][1];
    String plain="";
    for(int i=0;i < block;i++)
    {
        a[i][0]=b1.indexOfChar(cipher.charAt(i));
    }
    for(int i=0;i < block;i++)
    {
        for(int j=0;j < 1;j++)
        {
            for(int k=0;k < block;k++)
            {
                sum=sum+key[i][k]*a[k][j];
            }
            while(sum < 0)
            {
                sum+=26;
            }
        }
    }
    plain+=cipher.charAt(i);
}
}

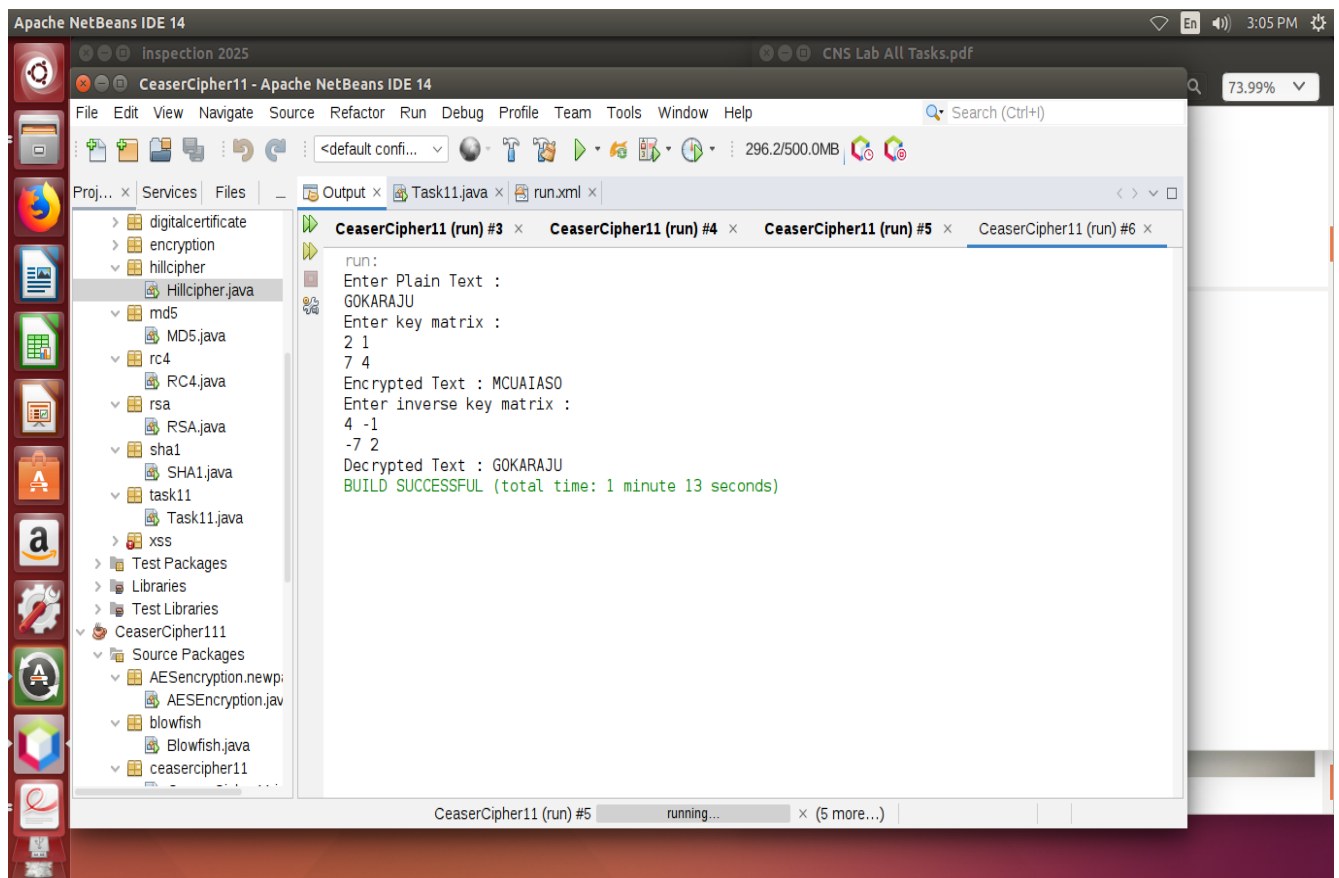
```

```

        }
        plainMatrix[i][j] = sum;
        sum = 0;
    }
    orf(int i=0;i < block;i++)
    {
        plain+=b1.charAtIndex(plainMatrix[i][0]);
    }
    return plain;
)
String Decrypt(String cipherText)throws Exception
{
    String plainText="";
    KeyInverseInsert();
    cipherText=cipherText.replaceAll(" ", "");
    cipherText=cipherText.toUpperCase();
    int len=cipherText.length();
    for(int i=0;i < len-1;i=i+block)
    {
        plainText+=decryptBlock(cipherText.substring(i,i+block));
        plainText+=" ";
    }
    return plainText;
}
}
class HillCipher
{
    public static void main(String args[])throws Exception
    {
        String plainText,cipherText;
        int block;
        Scanner scn=new Scanner(System.in);
        System.out.println("Enter plain-text:");
        plainText=scn.nextLine();
        System.out.println("Enter block size of matrix:");
        block=scn.nextInt();
        Hill hill=new Hill(block);
        plainText=plainText.replaceAll(" ", "");
        cipherText= hill.encrypt(plainText);
        System.out.println("Encrypted Text is:\n"+cipherText);
        String decryptedText= hill.Decrypt(cipherText);
        System.out.println("Decrypted Text is:\n"+decryptedText);
    }
}

```

OUTPUT:



VIVA QUESTIONS

1. Why does the key matrix in Hill Cipher need to be invertible in modular arithmetic?
2. How do you perform encryption and decryption using the Hill Cipher?
3. What are the advantages and limitations of the Hill Cipher compared to other classical ciphers?
4. How does matrix multiplication play a role in the Hill Cipher encryption process?